National University

of Computer and Emerging Sciences
Chiniot-Faisalabad Campus

Department of Artificial Intelligence and Data Science

Course Code	AI4002
Course Title	Computer Vision
Due Date	Sunday, 25 October 2026, 11:30 PM
Instructions:	

- Add group number in the file name.
- Only ONE student from each group will submit.
- Upload .pdf file and also submit printed hardcopy.
- Submit on Due Date. No Late Submission will be Tolerated.

Project Part 1 Submission

The submission should be prepared as one complete document containing the following sections:

1. Group Information

Provide a table containing:

Group No.	Student 1 Name	Roll No.	Student 2 Name	Roll No.	Project Title

2. Base Paper Summary

Provide a short paragraph or bullet-point summary of the selected Base Paper, including:

- Paper title
- Authors
- Publication year
- Journal name
- SCImago Journal Rank (SJR) Quartile
- Problem addressed
- Proposed model/approach
- Main reported results
- Key contribution of the paper

Upload the PDF of the Base Paper with the submission. Not Printed.

The Base Paper must satisfy the previously specified requirements: published within the last three years and in a Q1 or Q2 journal, with the quartile independently verified through SCImago.

3. Dataset Information

Provide a detailed table containing all relevant dataset information.

Information	Details
Dataset Name	
Dataset Link	
Source/Repository	
Number of Images	
Image Dimensions	
Image Format	
Classes/Labels	
Annotation Type	
Train/Validation/Test Split	
Modality	
Dataset Size	
License/Usage Information	

Students should also briefly explain any dataset preparation or basic preprocessing required before model training, consistent with the existing project requirements.

4. Experiment 1: Baseline

4.1 Model and Hyperparameters

Implement and retrain the Base Paper Model using the Original Dataset.

Provide a table containing the important experimental settings:

Hyperparameter / Setting	Value
Model Architecture	
Input Image Size	
Batch Size	
Number of Epochs	
Learning Rate	
Optimizer	
Loss Function	
Scheduler	
Train/Validation/Test Split	
Data Augmentation	
Hardware/GPU	

4.2 Results

Provide:

- Results table
- Appropriate evaluation metrics
- Training/validation graphs
- Loss curves
- Accuracy/metric curves where applicable
- Confusion matrix, segmentation visualization, detection visualization, or other relevant charts depending on the project type
- Difference with base paper results.

5. Experiment 2: Preprocessing Technique 1

5.1 Preprocessing Technique

State:

Preprocessing Technique:

Explain briefly:

- What technique was applied?
- How was it applied?
- Why was this technique selected for this dataset/problem?
- What improvement or effect was expected?

The preprocessing technique must be based on concepts covered during the first five weeks, such as intensity transformation, histogram equalization, histogram-based enhancement, Gaussian filtering, sharpening, or appropriate edge-based enhancement.

5.2 Experimental Settings

Keep the same Base Model and major hyperparameters as Experiment 1 wherever possible.

Provide the relevant hyperparameter table.

5.3 Results

Provide:

- Results table
- Appropriate evaluation metrics
- Graphs/charts
- Training/validation curves
- Relevant qualitative results/visualizations

6. Experiment 3: Preprocessing Technique 2

6.1 Preprocessing Technique

State:

Preprocessing Technique:

Explain:

- What technique was applied?
- How was it applied?
- Why was this technique selected?
- What improvement or effect was expected?

6.2 Experimental Settings

Provide the relevant hyperparameter/settings table.

The same Base Model should be used so that the primary experimental difference remains the preprocessing technique. This is required for a fair comparison.

6.3 Results

Provide:

- Results table
- Appropriate evaluation metrics
- Graphs/charts
- Training/validation curves
- Relevant qualitative results/visualizations

7. Comparison and Analysis

Compare all three experiments:

Metric	Experiment 1: Baseline	Experiment 2: Preprocessing 1	Experiment 3: Preprocessing 2
Metric 1			
Metric 2			
Metric 3			
Metric 4			

Then provide a detailed analysis:

- Which experiment performed better?
- Did Preprocessing 1 improve the baseline?
- Did Preprocessing 2 improve the baseline?
- Which preprocessing technique produced the largest improvement?

- Which metrics improved or decreased?
- Why might the preprocessing technique have produced this effect?
- Did preprocessing introduce any undesirable effects?
- What observations can be made from the visual results?
- What limitations were observed?

Do not simply state the highest score. Explain the observed behavior.

The existing project instructions already require students to discuss whether preprocessing improved, reduced, or otherwise changed model performance.

8. Proposed Advanced Direction for Part 2 of Project

This section is very important because it connects Part 1 with Part 2.

Based on the results of the three experiments, students should propose how they intend to improve upon the Base Paper Model in Part 2.

Discuss:

A. Observed Limitation

What limitation/problem was observed in the Base Model or current experiments?

B. Proposed Improvement

What advanced approach/model will be investigated?

C. Advanced Model 1

- Model/approach name
- Why selected?
- Expected advantage over the Base Model

D. Advanced Model 2

- Model/approach name
- Why selected?
- Expected advantage over the Base Model

E. Expected Contribution

Explain what they expect to investigate or discover through the advanced experiments.